QUINCY RYAN

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EXPERIENCE

CD International 2026–Present
Software Developer

- Built Mogent, a Go CLI that composes agent instructions and skills from reusable, versioned libraries, with drift detection across generated outputs, disk, and upstream sources.
- Designed advanced LLM agent tooling: multi-agent orchestration, shared skill libraries, and handoff/continuity workflows for long-running agent work.
- DevOps and Nix administration: flakes, pinned toolchains, reproducible dev shells, CI, and declarative NixOS / nix-darwin configuration across systems and repos.

Cognitive Control Lab, Dr. Xiaomo Chen Oct 2025–Present
Research Assistant *UC Davis Center for Neuroscience*

- Programmed and deployed a video-based eye-tracking and neurophysiology paradigm for non-human primates.
- Modeling eye-tracking data from meaning-map tasks with linear mixed-effects models (Bambi/PyMC).

Cognition in Context Lab, Dr. Yuko Munakata Jan 2025–Present
Research Assistant *UC Davis Center for Mind and Brain*

- Built and ran PsychoPy + EYNA eye-tracking experiments with adults and children; analysis in R and Python.

Computational Cognitive Neuroscience, Dr. Randall O'Reilly Jan 2025–Sept 2025
Programmer *CompCogNeuro / Cogent Core*

- Improved the UX of the textbook's neural network simulations in Go with the Cogent Core framework.

Janata Lab, Dr. Petr Janata Aug 2024–Apr 2025
Research Assistant, Sole Developer *UC Davis Center for Mind and Brain*

- Integrated, refactored, and extended multiple models of music cognition for the Brainmuse project (C++, Bash, Python, JACK audio).

Neuro-Tech Davis Nov 2024–Mar 2025
Project Manager *UC Davis*

- Led a team of 6 building an EEG attention-detection BCI with ML classification and adaptive refocusing stimuli.

PROJECTS

Vroca — Accessibility text-to-speech and assistive reading system: multi-engine ONNX synthesis, ASR word alignment, RSVP reading overlays, local agent input. Nix-packaged for Linux and macOS; staged Rust rewrite.
Exocortex — A family of Rust libraries and tools (graph-based todo system, memory, social simulation) under shared API, style, and toolchain standards.
nix-config — One declarative configuration for NixOS and nix-darwin machines: secrets (sops), dev environments, desktop.

EDUCATION

University of California, Davis *2023–2026*
B.S. Cognitive Science (Computational emphasis); minors in Neuroscience and Music. GPA 3.83
Phi Beta Kappa; Citation for Outstanding Performance in Cognitive Science (2025); Dean's List

Pasadena City College *2019–2023* **California Institute of the Arts**, Music Performance & Composition *2017–2019*

SKILLS

Languages: Rust, Go, Python, C, C++, R, MATLAB, Bash
Infrastructure: Nix / NixOS / nix-darwin, flakes, direnv, Git, GitHub Actions, Linux, macOS
AI & Agents: LLM agent orchestration, instruction/skill composition, ONNX inference, Bayesian modeling (PyMC)
Research: PsychoPy, eye tracking, EEG, mixed-effects modeling, LaTeX